PrimaryAI — AI Engine Architecture

How a lesson pack is generated from a teacher request to a finished, curriculum-aligned output

TEACHER REQUEST

Year group · Subject · Topic · Teacher profile (ability mix, SEND focus, EAL %, teaching approach) · Lesson context (unit position, AfL note, today's adjustments) · Uploaded file context

CACHE LOOKUP

Layer 1 of 2

In-memory cache checked first (sub-ms). On cold start or miss: persistent Supabase cache checked (24h TTL, shared across all Cloud Run instances).

⚡ Cache hit → return immediately, skip all AI calls

CACHE MISS

CURRICULUM RETRIEVAL

Retrieves up to 5 UK National Curriculum objectives for this year group + subject + topic. Uses Cloudflare D1 vector search (cosine similarity on BGE embeddings) with keyword-match fallback. These objectives anchor the entire generation.

GENERATION PASS — TIERED MULTI-PROVIDER ENSEMBLE

- **TIER 1** — Gemini 2.5 Flash · Cerebras Llama-70B · Groq Llama-70B
- **TIER 2** — OpenRouter Llama-70B · Mistral Small · Cohere Command-R+
- **TIER 3** — Cloudflare Llama-8B · HuggingFace Mistral-7B

✓ Tier 1 fires in parallel. If 2+ succeed → ensemble from Tier 1 only. Stop.

~ If only 1 Tier 1 succeeds → also run top 2 from Tier 2 for ensemble enrichment.

↓ If all Tier 1 fail → run all of Tier 2. Rate-limited providers are skipped automatically (5 min cooldown).

↓ Last resort: Tier 3 small models. Result is lower quality but never a silent failure.

SMART ENSEMBLE + SCORING

All valid responses are scored independently, then the best fields are assembled across providers:

Learning Objectives	Activities	Mini Assessment
Highest curriculum objective coverage score	Deepest differentiation across support/expected/GD	Most complete 4-question ascending-challenge set

+ objective coverage ×5

+ numbered steps in worked example ×2

+ field richness (length >80 chars)

– vague language ("various", "etc")

– support ≈ greater depth (too similar)

QUALITY PASS

Single provider

Fastest available provider reviews the draft against 11 criteria (objectives start with Bloom's verb, worked example shows all steps, greater depth demands reasoning, SEND adaptations name specific strategies, etc.). If rejected: only the flagged sections are regenerated — not the whole pack.

ALIGNMENT PASS

Single provider

Checks curriculum objective coverage ratio (≥ 60% threshold). If below threshold: fastest provider realigns objectives, teacher explanation, worked

example, and activities to the retrieved NC objectives. Skipped entirely if already aligned.

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FINALIZE

Gap-fill with sensible UK defaults · UK spelling normalisation (colour, organise, behaviour, centre, analyse) · Zod schema validation · Programmatic slide generation · Confidence scoring (high / medium / low based on objectives retrieved + context used)

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CACHE WRITE

Result written to in-memory cache immediately. Also written to Supabase engine_cache table asynchronously (non-blocking). Next request for the same topic + profile hits the cache — zero AI calls.

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LESSON PACK OUTPUT

Learning objectives · Teacher + pupil explanations · Worked example · Lesson hook · Timing guide · Differentiated activities (Support / Expected / Greater Depth) · Differentiation by group (Lower / Core / Higher) with Bloom's-levelled questions · Critical thinking questions (all 6 Bloom's levels) · SEND adaptations · Safety notes · Plenary · Mini assessment · Slides · Rationale tags · Next steps

8	3–5	24h	3
AI Providers	API calls per generation (was 24+)	Persistent cache TTL	Quality passes